Double-click (or enter) to edit

```
from google.colab import files

uploaded = files.upload()
```

Choose Files placement_…Dataset.csv

placement_predict_50k Dataset.csv(text/csv) - 6811359 bytes, last modified: 3/8/2026 - 100% done

```
Saving placement_predict_50k Dataset.csv to placement_predict_50k Dataset.csv
```

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from matplotlib.backends.backend_pdf import PdfPages

sns.set_style("whitegrid")   # optional: gives cleaner-looking charts
```

```
df = pd.read_csv("placement_predict_50k Dataset.csv")
```

```
print("First 5 rows of the dataset:")
display(df.head())          # use display() in Colab; use print(df.head()) elsewhere
```

```
First 5 rows of the dataset:
```

	StudentID	Gender	City	CollegeTier	Stream	Specialisation	Hostel	HistoryOfBacklogs	SGP
0	1	Male	Ahmedabad	Tier2	ECE	Networking	No	No	
1	2	Female	Mumbai	Tier2	ECE	DataScience	Yes	Yes	
2	3	Male	Kolkata	Tier2	IT	DataScience	Yes	No	
3	4	Male	Jaipur	Tier1	CS	AI	No	No	
4	5	Male	Pune	Tier2	IT	DataScience	Yes	No	

5 rows × 32 columns

```
print("\nShape of dataset (rows, columns):", df.shape)

print("\nColumn names:")
print(df.columns.tolist())

print("\nData types and non-null counts:")
df.info()

print("\nMissing values per column:")
print(df.isnull().sum())
```

```
Shape of dataset (rows, columns): (50000, 32)

Column names:
['StudentID', 'Gender', 'City', 'CollegeTier', 'Stream', 'Specialisation', 'Hostel', 'HistoryOfBack

Data types and non-null counts:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 32 columns):
 #   Column              Non-Null Count  Dtype
---  ------              --------------  -----
 0   StudentID           50000 non-null  int64
 1   Gender              50000 non-null  object
 2   City                50000 non-null  object
```

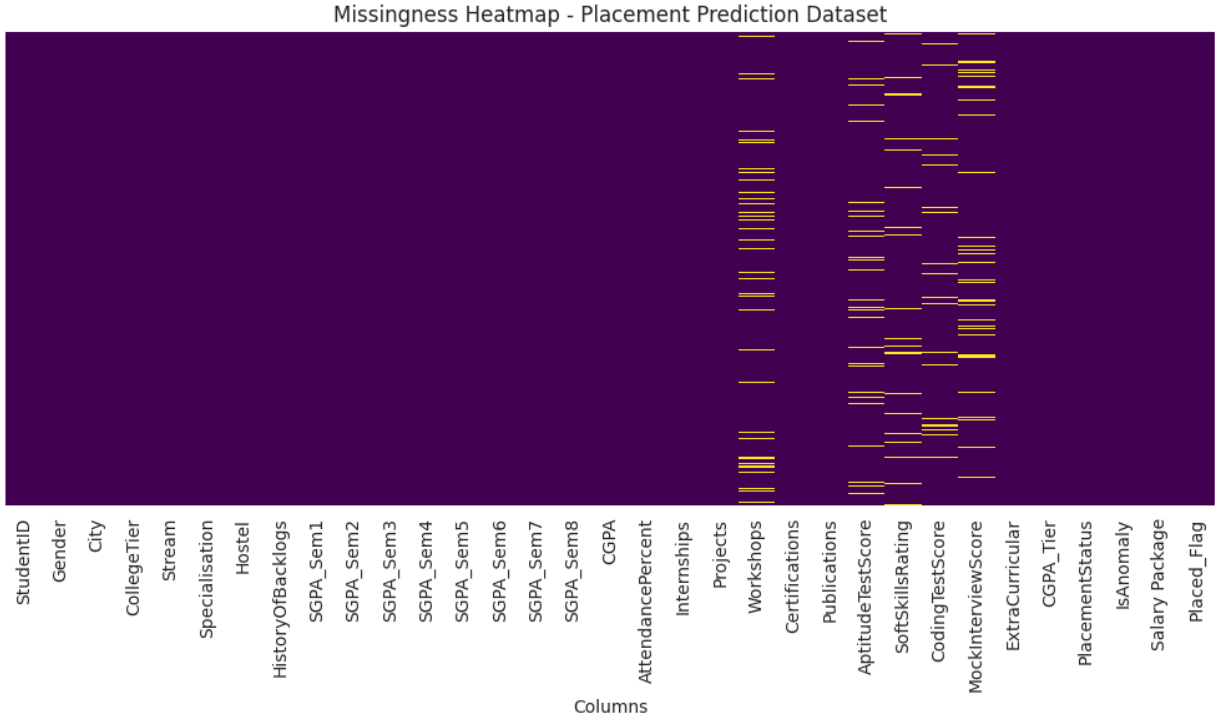
```
 3   CollegeTier          50000 non-null  object
 4   Stream               50000 non-null  object
 5   Specialisation       50000 non-null  object
 6   Hostel               50000 non-null  object
 7   HistoryOfBacklogs    50000 non-null  object
 8   SGPA_Sem1            50000 non-null  float64
 9   SGPA_Sem2            50000 non-null  float64
 10  SGPA_Sem3            50000 non-null  float64
 11  SGPA_Sem4            50000 non-null  float64
 12  SGPA_Sem5            50000 non-null  float64
 13  SGPA_Sem6            50000 non-null  float64
 14  SGPA_Sem7            50000 non-null  float64
 15  SGPA_Sem8            50000 non-null  float64
 16  CGPA                 50000 non-null  float64
 17  AttendancePercent    50000 non-null  float64
 18  Internships          50000 non-null  int64
 19  Projects             50000 non-null  int64
 20  Workshops            45512 non-null  float64
 21  Certifications       50000 non-null  int64
 22  Publications         50000 non-null  int64
 23  AptitudeTestScore    45971 non-null  float64
 24  SoftSkillsRating     46474 non-null  float64
 25  CodingTestScore      47024 non-null  float64
 26  MockInterviewScore   45043 non-null  float64
 27  ExtraCurricular      50000 non-null  int64
 28  CGPA_Tier            50000 non-null  object
 29  PlacementStatus      50000 non-null  int64
 30  IsAnomaly            50000 non-null  int64
 31  Salary Package       50000 non-null  float64
dtypes: float64(16), int64(8), object(8)
memory usage: 12.2+ MB

Missing values per column:
StudentID                0
Gender                   0
City                     0
CollegeTier              0
Stream                   0
Specialisation           0
Hostel                   0
HistoryOfBacklogs        0
SGPA_Sem1                0
```

```
df['Placed_Flag'] = df['PlacementStatus'].apply(lambda x: 1 if x == 'Placed' else 0)
```

```
num_cols = ['CGPA', 'AttendancePercent', 'AptitudeTestScore',
            'CodingTestScore', 'SoftSkillsRating', 'Salary Package']
```

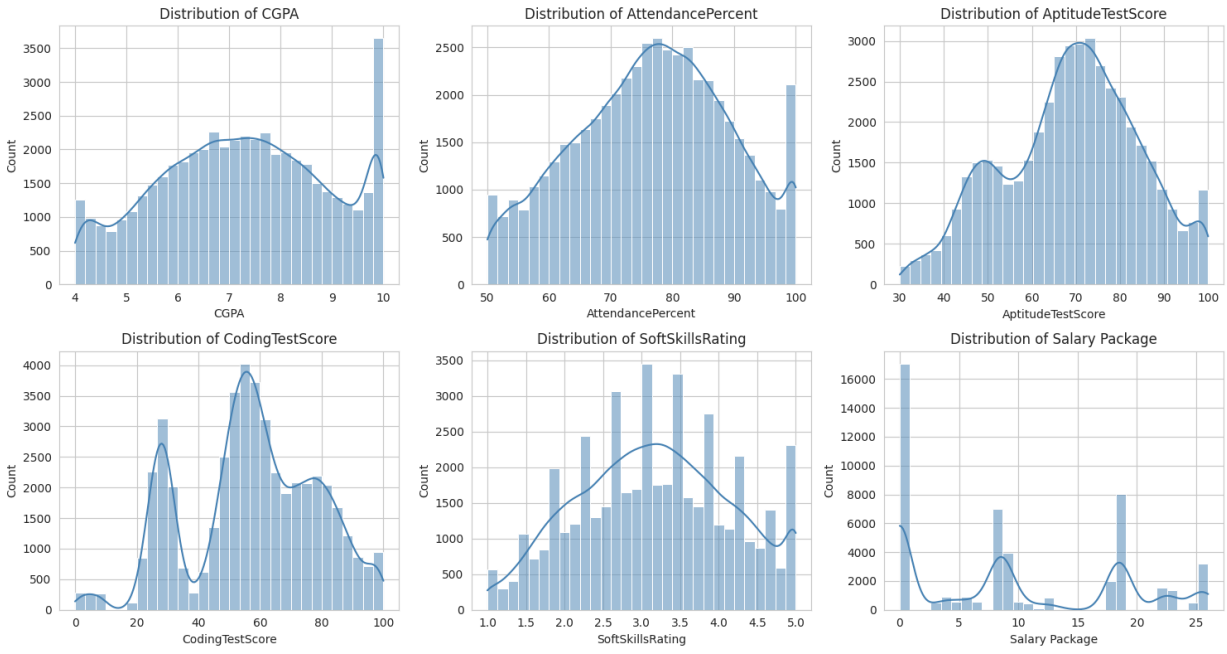
```
fig1 = plt.figure(figsize=(10, 6))
sns.heatmap(df.isnull(), cbar=False, cmap="viridis", yticklabels=False)
plt.title("Missingness Heatmap - Placement Prediction Dataset")
plt.xlabel("Columns")
plt.tight_layout()
plt.show()
```




```
fig2, axes = plt.subplots(2, 3, figsize=(15, 8))

for i, col in enumerate(num_cols):
    row, colpos = divmod(i, 3)
    sns.histplot(df[col].dropna(), bins=30, kde=True,
                 ax=axes[row, colpos], color='steelblue')
    axes[row, colpos].set_title(f'Distribution of {col}')

plt.tight_layout()
plt.show()
```




```
fig3 = plt.figure(figsize=(10, 6))

placement_rate = (
    df.groupby('Stream')['Placed_Flag'].mean().sort_values(ascending=False) * 100
)
```

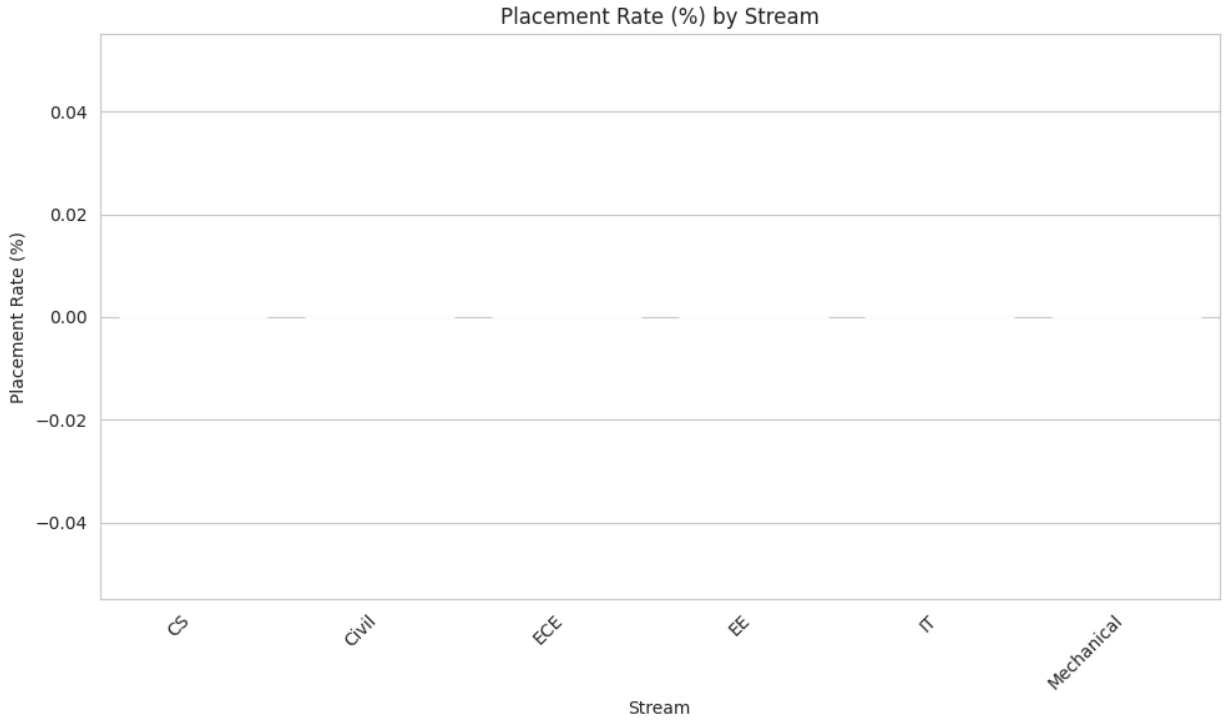
```
sns.barplot(x=placement_rate.index, y=placement_rate.values, palette='crest')
plt.title('Placement Rate (%) by Stream')
plt.xlabel('Stream')
plt.ylabel('Placement Rate (%)')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()

print("\nPlacement rate (%) by stream:")
print(placement_rate.round(2))
```

```
/tmp/ipykernel_1180/2158546150.py:7: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `

  sns.barplot(x=placement_rate.index, y=placement_rate.values, palette='crest')
```

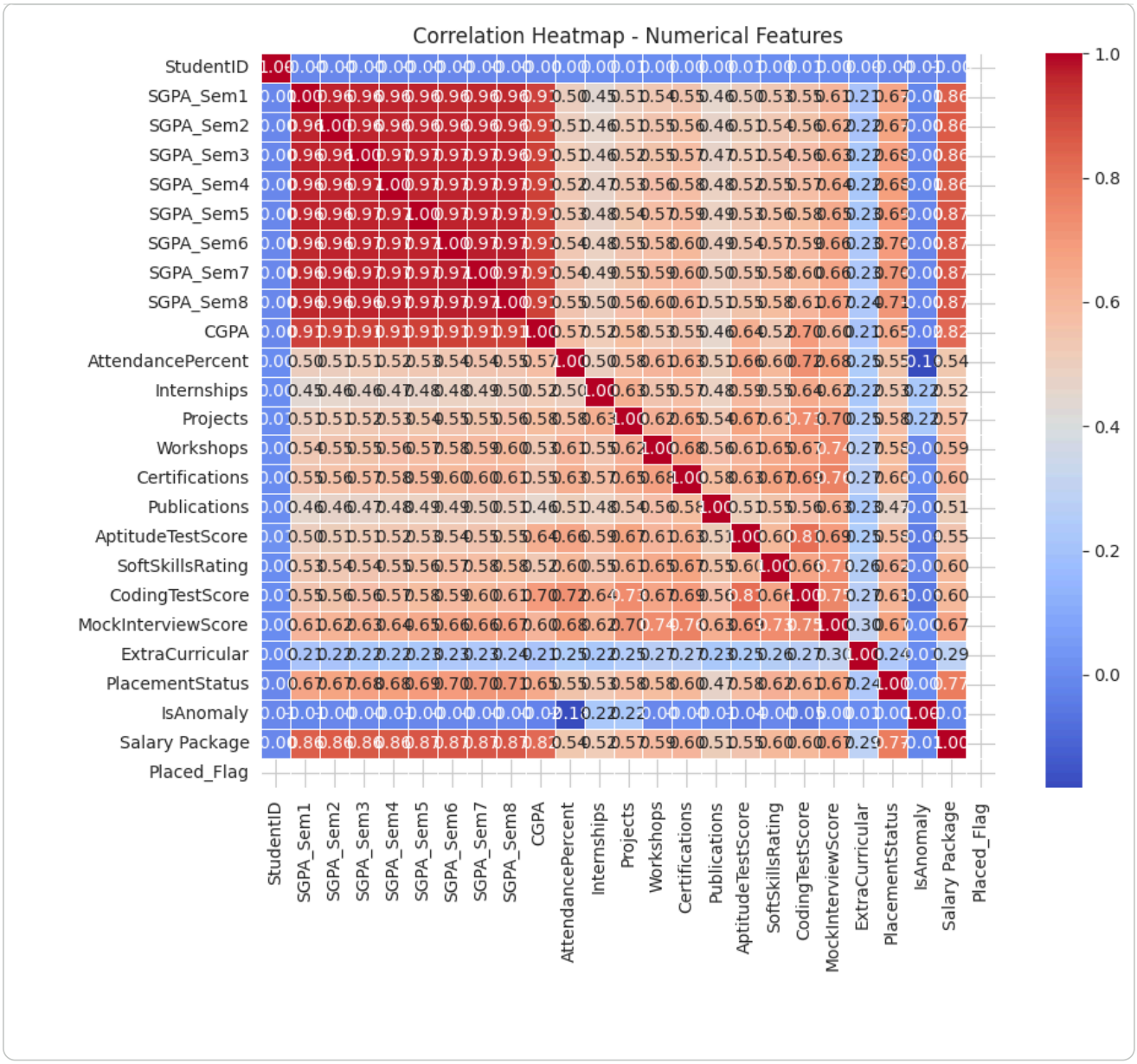



```
Placement rate (%) by stream:
Stream
CS            0.0
Civil         0.0
ECE           0.0
EE            0.0
IT            0.0
Mechanical    0.0
Name: Placed_Flag, dtype: float64
```

```
fig4 = plt.figure(figsize=(10, 8))

numeric_df = df.select_dtypes(include=['int64', 'float64'])
corr_matrix = numeric_df.corr()

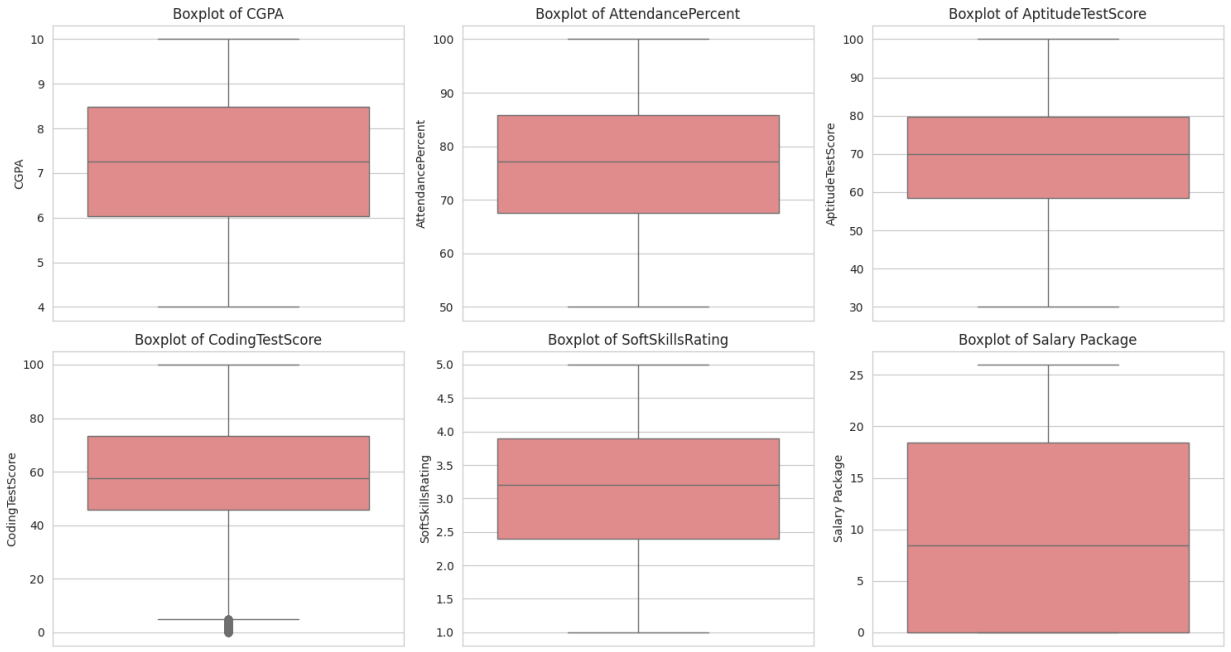
sns.heatmap(corr_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            square=True, linewidths=0.5)
plt.title('Correlation Heatmap - Numerical Features')
plt.tight_layout()
plt.show()
```




```
fig5, axes5 = plt.subplots(2, 3, figsize=(15, 8))

for i, col in enumerate(num_cols):
    row, colpos = divmod(i, 3)
    sns.boxplot(y=df[col], ax=axes5[row, colpos], color='lightcoral')
    axes5[row, colpos].set_title(f'Boxplot of {col}')

plt.tight_layout()
plt.show()
```




```
pdf = PdfPages('EDA_Report.pdf')

pdf.savefig(fig1)   # Page 1: Missingness Heatmap
pdf.savefig(fig2)   # Page 2: Six Histograms
pdf.savefig(fig3)   # Page 3: Placement Rate by Stream
pdf.savefig(fig4)   # Page 4: Correlation Heatmap
pdf.savefig(fig5)   # Page 5: Boxplots
```